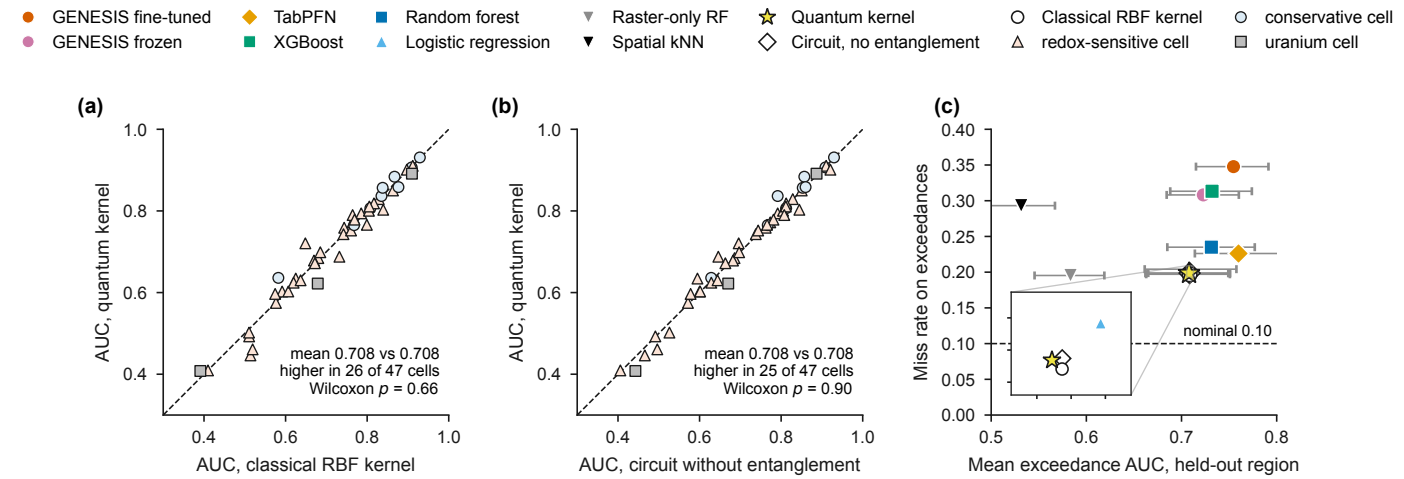
GENESIS fine-tuned
GENESIS frozen
TabPFN
XGBoost
Random forest
Logistic regression
Raster-only RF
Spatial kNN
Quantum kernel
Circuit, no entanglement
Classical RBF kernel
redox-sensitive cell
conservative cell
uranium cell
(a)
AUC, quantum kernel
AUC, classical RBF kernel
mean 0.708 vs 0.708
higher in 26 of 47 cells
Wilcoxon p = 0.66
(b)
AUC, quantum kernel
AUC, circuit without entanglement
mean 0.708 vs 0.708
higher in 25 of 47 cells
Wilcoxon p = 0.90
(c)
Miss rate on exceedances
Mean exceedance AUC, held-out region
nominal 0.10